



MINING DEALER BEST PRACTICE

Rolling Maintenance Parts for Reducing PM Time 794/798 AC

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Rolling Maintenance Parts for Reducing PM Time	
794/798 AC	1
1.0 Introduction.....	2
2.0 Best Practice Description	2
3.0 Implementation Steps.....	4
4.0 Benefits	5
5.0 Resources Required.....	5
6.0 Supporting Attachments / References	5
7.0 Related Best Practices	7
8.0 Acknowledgements	7

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September 2024
1.01-240905-1

1.0 Introduction

Taking time during a PM to inspect and clean reusable screens, filters and magnetic plugs can directly impact maintenance time. Utilizing a set of rotating maintenance parts, such as clean screens or a Centrifugal COF with new seals during the PM and cleaning the parts after the truck has been returned to service and before the next PM, will reduce the PM timeline and increase availability of a technician during machine downtime.

2.0 Best Practice Description

To reduce the PM timeline, it's necessary to have additional parts so that the removed parts can be cleaned, analyzed and potentially repaired post PM. Those parts are proposed on the next table (1). Rolling these parts reduce 3.5 hours on every PM.

Item	Description	System	Parts Number	Time reduced hours
1	Screens	Drive Train	313-4082**	1
2	COF Rotor AS Filter Centrifugal	Engine lubrication	494-6433***	1
3	Oil Screen	Engine lubrication	364-8521**	0.5
4	Magnetic plugs	Final drive , Wheel , Drive Pump , and High pressure pump	Wheels (6V-3298) ** Final drive (6V-0439)** High pressure fuel pump (3J-1907) ** Driven Pump (377- 9588)**	1
			Total	3.5

** For 794 and 798 AC all parts are used on both trucks

***Only engines with Centrifugal oil filter.

Item 1 - Changing 7 clean screens during the PM only takes 30 min. The other time that it takes for cleaning with water will take an hour.

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			NUMBER 1.01-240905-1



Image 2 Clean Screen Ready to install



Image 3: Clean Screens installed on the inverter cabinet

Item 2 - Rolling COF with new seals and paper ready to change, it's only necessary to unbolt two clamps and change the group part



Image 4: COF Centrifugal Oil Filter ready to install

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Rolling Maintenance Parts for Reducing PM Time 794 / 798 AC	DATE	CHG NO	NUMBER
	9/5/2024	00	1.01-240905-1

Item 3 - Clean Oil Screen ready to change, also gives the availability for change if there are problems on the screen, having time to replace it for the next PM with another new one.



Image 5: Oil Screen

Item 4 - Magnetic Plugs, having additional magnetic plugs inside box, waiting for the next PM. Improves exchange time and allows opportunity to properly analyze & trend magnetic plug results.



Image 6 Wood Box with plugs

3.0 Implementation Steps

1. Train and inform about the PM technical personnel about the rolling parts that must be ready for the maintenance
2. Every 6 month evaluate the rolling parts to assess if some parts are worn or have any defects that could affect the performance
3. Send magnetic plugs to CM analyst to inspect & record results, creating a history of weight, color, etc.

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Rolling Maintenance Parts for Reducing PM Time 794 / 798 AC	DATE 9/5/2024	CHG NO 00	NUMBER 1.01-240905-1

DATOS DEL EQUIPO		DATOS DEL MANTENIMIENTO	
EQUIPO	CA-6	TIPO DE PM	280H
HOROMETRO	3329	FECHA	02-04-19
RUEDA DERECHA		RUEDA IZQUIERDA	
			

4.0 Benefits

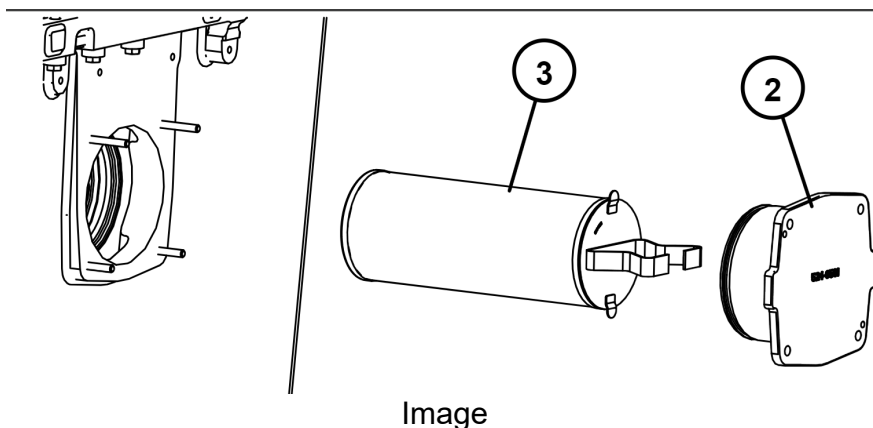
- Reduce maintenance time
- Provide better control for the rolling parts to reduce failures.
- Having better control of magnetic plugs, to study and create history.

5.0 Resources Required

- Necessary to have a kit to change as rolling part, described on table 1
- A dedicated area and process to weigh and analyze metal generated by magnetic plugs

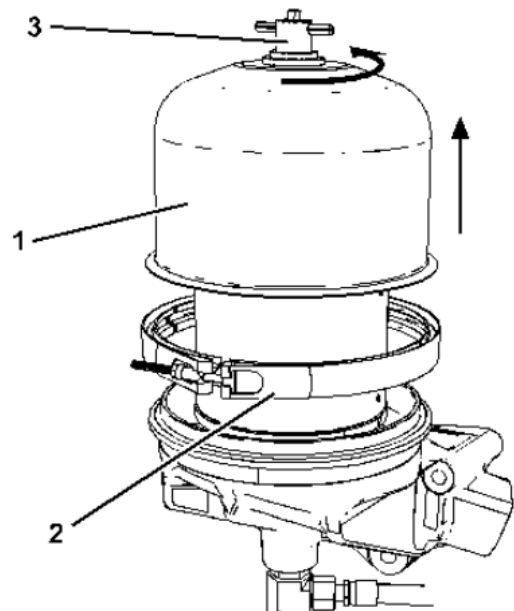
6.0 Supporting Attachments / References

Every 500 hours, Engine Oil pan sump screen (item 3) must be inspected and clean (M0106737-03).



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			NUMBER 1.01-240905-1

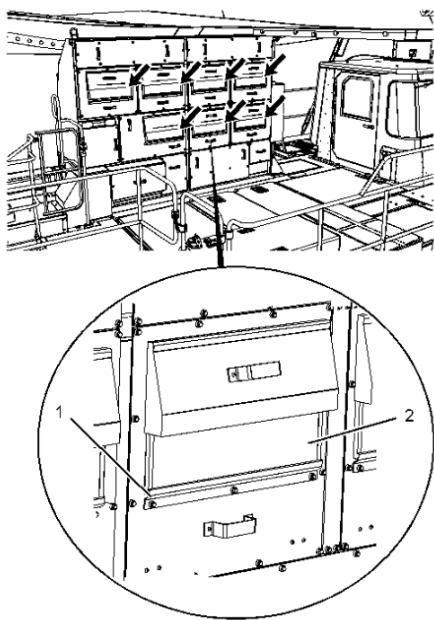
Every 500, it is necessary to clean the centrifugal oil filter (M0106737-03).



Only is needed to disassembly an assembly item 1 in the image. Reusing the clamps 3 and 2.

Paper COF and seals are new and installed before the PM

Every 500 hours, it is necessary to clean / inspect the air filter element of electric cabinet (metal sand filters - M0106737-03).



Clean the screens with water (low pressure) and wait all the filters to dry completely.

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Rolling Maintenance Parts for Reducing PM Time 794 / 798 AC	DATE	CHG NO	NUMBER
	9/5/2024	00	1.01-240905-1

The implementation will cost \$7190.84 one kit for 4 trucks to use the same parts from one truck to another. With more trucks it will be necessary to increase by two or more kits.

Item	Description	System	Parts Number	Cost (Reference) \$ Sales price
1	Screens	Drive Train	313-4082**	56.70
2	COF Rotor AS Filter Centrifugal	Engine lubrication	494-6433***	6345.39
3	Oil Screen	Engine lubrication	364-8521**	439.80
4	Magnetic plugs	Power train	Wheels (6V-3298) **	223.95
5		Power train	Final drive (6V-0439) **	68.92
6		Engine	High pressure fuel pump (3J-1907) **	2.23
7		Engine	Driven Pump (377-9588)**	53.85
			Total	7190.84

7.0 Related Best Practices

0509-1.02-1167 – Magnetic Plug Inspection Trays and Inspection Process

8.0 Acknowledgements

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